



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2023 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A

4045/02

Paper 2

31 July 2023

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Section A

Answer **all** questions.

Section B

Answer **one** question.

The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total number of marks for this paper is 70.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator or 3.142.

Total Marks

Mathematical Formulae*Compound interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f} \right)^2}$$

Section A (62 marks)

Answer **all** the questions in this section.

1 Written as the product of its prime factors, $504 = 2^3 \times 3^2 \times 7$.

(a) Write 2100 as the product of its prime factors.

Answer [2]

(b) Giving each answer as the product of its prime factors,

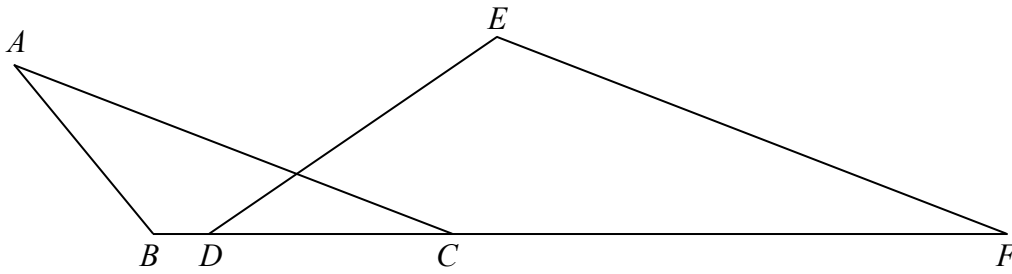
(i) find the lowest common multiple (LCM) of 504 and 2100.

Answer [1]

(ii) find the smallest positive integer value of m such that $\frac{504}{m}$ is a perfect cube.

Answer $m =$ [1]

2



Triangle ABC is similar to triangle DEF .

- (a) Identify all the pairs of angles in triangle ABC and triangle DEF that are equal.

Answer

[2]

- (b) $AC = 8.1$ cm, $DF = 14.175$ cm and $EF = 9.45$ cm.

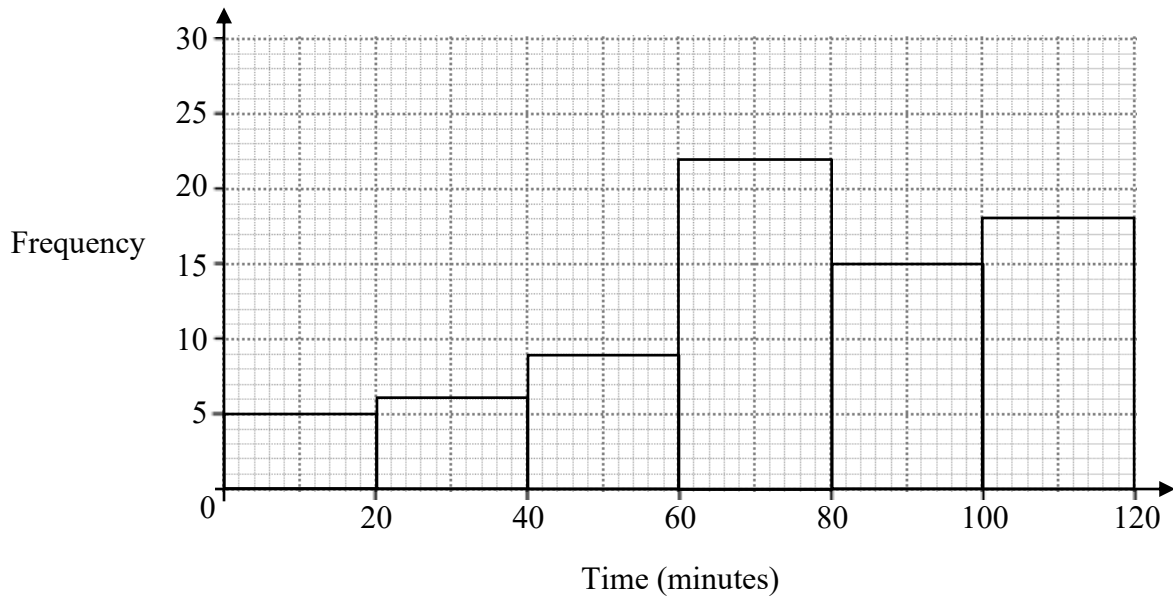
- (i) Calculate the length of BC .

Answer $BC = \dots\dots\dots$ cm [2]

- (ii) Find the scale factor for the reduction of triangle DEF to triangle ABC .

Answer $\dots\dots\dots$ [1]

- 3 A group of adults took part in a sailboat race.
 The time limit for this race is 120 minutes.
 The histogram below shows the distribution of the times taken by these adults to complete the race.



Mac claims that 24% of these adults who took part in the race reached the time limit.

Explain why he might not be correct.

Answer

.....

..... [2]

4 (a) y is directly proportional to the square of x .

(i) Given that $y = 21$ when $x = 6$, write a formula for y in terms of x .

Answer [2]

(ii) Find y when $x = 9$.

Answer $y =$ [1]

(b) Solve these simultaneous equations.

$$\begin{aligned} 2x + y &= 1 \\ 3x - 4y &= 29 \end{aligned}$$

Show your working.

Answer $x =$

$y =$ [3]

(c) Solve $\frac{3x+1}{4} - \frac{9-x}{5} = 7$.

Answer $x = \dots\dots\dots$ [3]

- 5 (a) Expand and simplify $2h(3h+5) - h(1+4h)$.

Answer [2]

- (b) Factorise completely $72m^2 - 32n^2$.

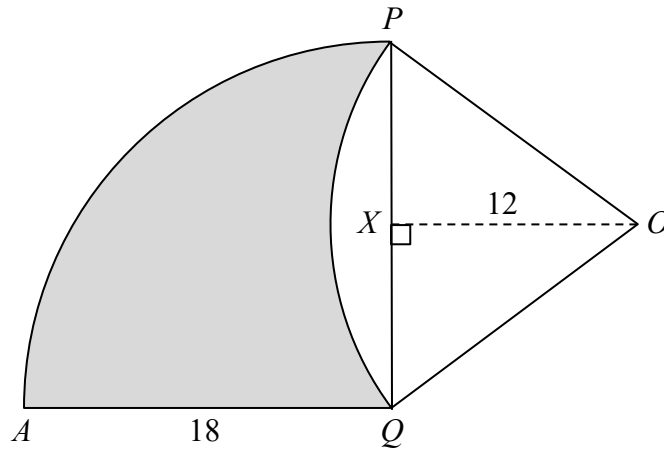
Answer [2]

- (c) Rearrange this formula to make a the subject.

$$b^3 = \frac{2a}{a-1}$$

Answer $a =$ [3]

6



The diagram shows a quarter of a circle, QAP , with radius 18 cm and centre Q .
 OQP is a sector of another circle with centre O .
 $OX = 12$ cm and angle $OXQ = 90^\circ$.

Calculate

(a) OQ ,

Answer $OQ = \dots\dots\dots$ cm [2]

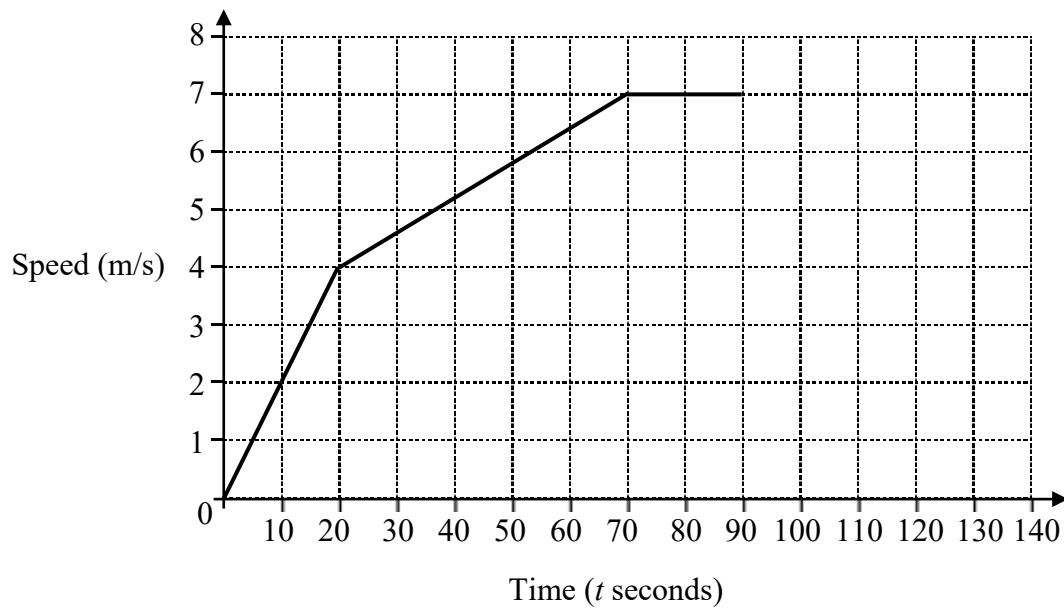
(b) angle POQ ,

Answer angle $POQ = \dots\dots\dots$ [2]

(c) the perimeter of the shaded region.

Answer $\dots\dots\dots$ cm [4]

- 7 The graph shows part of the speed-time graph of Lina's cycle journey.



- (a) Describe the motion between $t = 70$ and $t = 90$.

Answer

.....

.....[1]

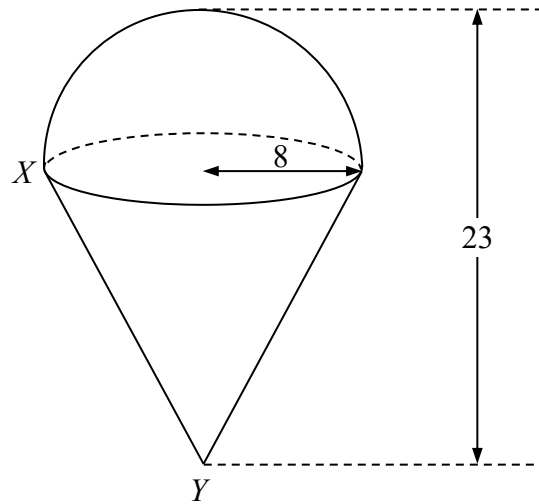
- (b) Find the acceleration for the first 20 seconds of Lina's journey.

Answer m/s^2 [1]

- (c) After 90 seconds, Lina slowed down to a stop with constant deceleration. She travelled a further 42 m during the deceleration.

Given that the area under the graph represents the total distance travelled, find the value of t when Lina stopped.

Answer $t = \dots\dots\dots$ [2]



The diagram shows a solid made from a hemisphere and a cone.
 The hemisphere and cone each have a radius of 8 cm.
 The total height of the solid is 23 cm.

- (a) Calculate the volume of the solid.

Answer cm^3 [3]

- (b) Find the slant height, XY , of the cone.

Answer $XY =$ cm [1]

- (c) Calculate the total surface area of the solid.

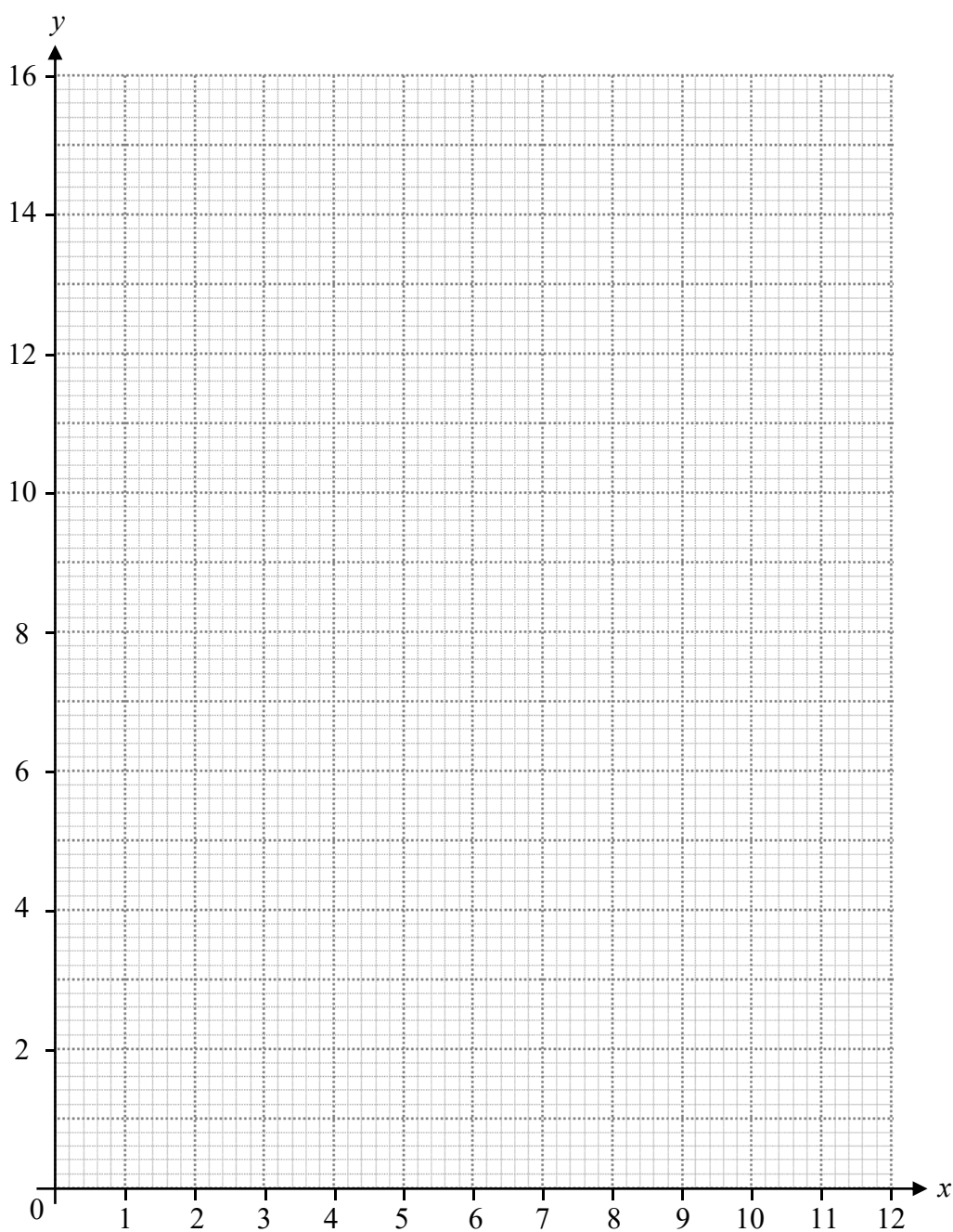
Answer cm^2 [3]

- 9 (a) Complete the table of values for $y = x + \frac{15}{x}$.

x	1	2	3	4	5	6	7	8
y	16	9.5	8	7.75	8		9.14	9.88

[1]

- (b) Draw the graph of $y = x + \frac{15}{x}$ for $1 \leq x \leq 8$.



[3]

- (c) Use your graph to find the largest value of x when $y = 9$.

Answer $x = \dots\dots\dots$ [1]

- (d) By drawing a suitable tangent, find the gradient of the graph when $x = 2$.

Answer $\dots\dots\dots$ [2]

- 10 (a)** Royal Bank offers a savings account with a rate of compound interest of 2.65% per year. Wayne invests \$11 200 in this account. He leaves the money in the account for 3 years.

Calculate the total amount of interest he has earned after 3 years.
Give your answer correct to the nearest cent.

Answer \$ [3]

- (b)** It costs a total of \$21 000 to renovate Wayne's bathroom and kitchen.

To pay for this cost, Wayne is thinking of borrowing some money from a bank by applying for a personal loan.

To ensure that Wayne is able to pay this loan, after receiving the money from his investment in Royal Bank, he decided that the interest to be paid on this loan must **not** exceed the interest earned from his investment in part **(a)**.

The table below shows some information that Wayne found about personal loans from three banks.

	Alpha Bank	Brown Bank	Central Bank
Eligibility Criteria: Minimum Annual Income	\$30 000	\$45 000	\$20 000
Minimum Loan Amount	\$1000	\$3000	\$1000
Maximum Loan Amount	Up to 6 times of monthly income	Up to 6 times of monthly income	Up to 8 times of monthly income
Loan Period	6, 12, 24, 36, 48 or 60 months	6, 12, 24, 36, 48 or 60 months	1, 2, 3, 4 or 5 years
Simple Interest Rate per annum	3.87%	3.87%	3.48%

- (i) If Wayne loans from Central Bank for a period of 1 year, calculate the maximum amount he can loan, such that the interest to be paid for this loan does not exceed the interest earned from his investment in part (a).
Give your answer correct to the nearest dollar.

Answer \$ [2]

- (ii) Wayne's monthly income is \$3550.
He wants to loan as much as possible to pay for the cost of renovating his bathroom and kitchen.

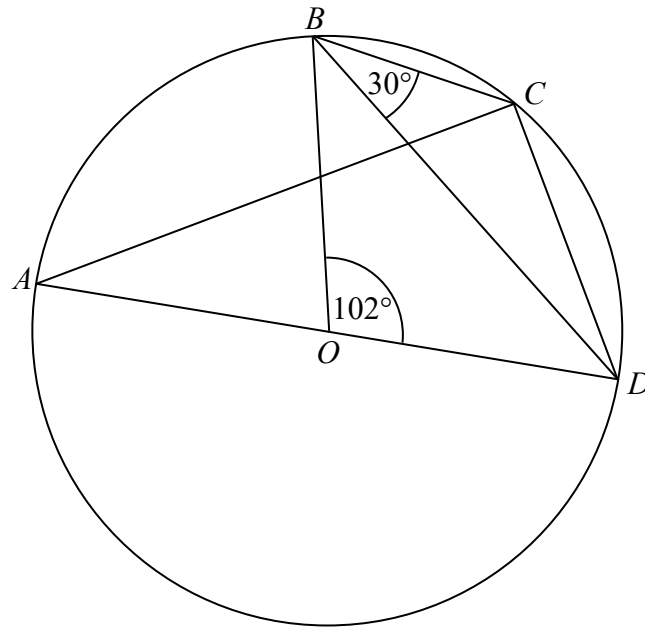
Which bank, Alpha Bank, Brown Bank or Central Bank, should Wayne loan from and what loan period should he get to pay the least amount of interest?
Justify the decision you make and show your calculations clearly.

Answer

Section B (8 marks)

Answer **one** question from this section. Each question carries 8 marks.

11 (a)



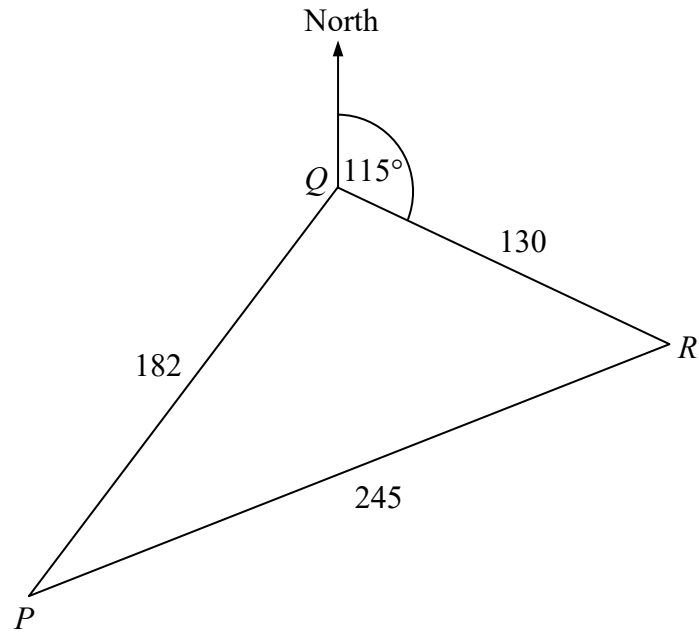
A, B, C and D are points on the circumference of a circle, centre O .
Angle $BOD = 102^\circ$ and angle $CBD = 30^\circ$.

Complete these statements by calculating the size of each angle.
Give a reason for each statement.

Statement	Reason
Angle $CAD = \dots\dots\dots$	$\dots\dots\dots$
	$\dots\dots\dots$
	$\dots\dots\dots$
Angle $BCD = \dots\dots\dots$	$\dots\dots\dots$
	$\dots\dots\dots$
	$\dots\dots\dots$

[2]

(b)



A boat travels from P to Q to R and then back to P again.

$PQ = 182$ m, $QR = 130$ m and $PR = 245$ m.

The bearing of R from Q is 115° .

(i) Calculate the bearing of P from Q .

Answer [4]

(ii) At P , there is a lighthouse of height 45 m.

Calculate the angle of elevation of the top of the lighthouse from Q .

Answer [2]

12 The table summarises the masses of 130 mangoes at Farm A.

Mass (x g)	$500 < x \leq 600$	$600 < x \leq 700$	$700 < x \leq 800$	$800 < x \leq 900$	$900 < x \leq 1000$
Number of mangoes (frequency)	11	15	26	45	33

- (a) (i)** Calculate an estimate of the mean and standard deviation of these masses.

Answer Mean g

Standard deviation g [3]

- (ii)** At Farm B, the mean mass of 130 mangoes is 855 g and the standard deviation is 134 g.

Which farm, A or B, has mangoes with more consistent masses?
Give a reason for your answer.

Answer

.....

.....[1]

- (b) Two of the 130 mangoes at Farm A are picked at random.

Calculate the probability that

- (i) both mangoes have a mass of more than 800 g,

Answer [2]

- (ii) at least one of the mangoes have a mass of at most 700 g.

Answer [2]

END OF PAPER

